

PERM, Russian Federation

WMO: 282240

Lat: 57.9925N

Lon: 56.3356E

Elev: 561

StdP: 14.40

Time Zone: 5.00 (E05)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.3	-14.5	-26.0	1.3	-20.1	-19.3	1.9	-14.0	16.5	19.2	15.1	19.3	2.6	10	0.579

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	85.2	68.4	82.1	66.6	78.6	64.6	70.2	82.1	68.4	79.0	66.7	75.8	5.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	98.3	75.0	64.5	92.8	73.2	62.9	87.6	72.2	34.4	82.6	33.0	79.0	31.6	75.8	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.4	12.7	11.5	DB	-27.1	90.2	8.8	3.3	-33.5	92.6	-38.7	94.5	-43.6	96.4	-50.0	98.8	(4)
(5)			WB	-27.3	72.3	8.8	1.8	-33.7	73.6	-38.8	74.7	-43.8	75.7	-50.2	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.8	8.2	11.0	24.8	39.5	53.0	61.1	65.1	61.6	51.0	37.7	24.7	13.9		
	DBStd	22.15	13.00	11.77	8.70	8.15	9.36	8.03	7.75	7.62	7.37	8.31	10.83	13.00		
	HDD50	5934	1295	1093	781	330	79	11	1	3	73	391	759	1118		
	HDD65	10153	1760	1513	1246	765	387	165	98	159	422	846	1209	1583		
	CDD50	1477	0	0	0	15	171	346	470	362	104	9	0	0		
	CDD65	221	0	0	0	1	14	49	102	52	3	0	0	0		
	CDH74	1970	0	0	0	2	176	423	936	418	15	0	0	0		
	CDH80	519	0	0	0	0	26	93	284	116	0	0	0	0		
	WSAvg	5.4	5.6	5.3	6.2	6.1	5.7	4.8	4.4	4.2	4.9	5.8	6.0	5.7		
	PrecAvg	24.2	1.5	1.0	1.1	1.4	2.1	2.8	3.1	2.9	2.5	2.4	1.9	1.5		
(15) Precipitation	PrecMax	34.4	2.8	2.8	3.0	3.6	5.9	6.8	6.2	6.9	5.4	4.0	3.4	2.6		
	PrecMin	15.7	0.4	0.2	0.0	0.1	0.6	0.4	0.6	0.9	0.5	0.4	0.7	0.1		
	PrecStd	4.8	0.6	0.6	0.7	0.8	1.3	1.7	1.5	1.5	1.3	0.9	0.8	0.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	34.9	49.6	71.4	84.1	86.4	89.6	89.8	76.1	63.2	48.6	36.2		
		MCWB	32.7	32.5	40.5	55.5	62.6	66.3	69.0	70.9	60.7	50.7	46.2	35.4		
	2%	DB	32.4	33.1	43.7	62.4	79.1	82.6	86.2	82.4	70.1	57.3	44.5	34.1		
		MCWB	31.4	31.3	36.3	50.6	60.0	65.9	69.9	68.3	58.4	49.5	42.3	33.0		
	5%	DB	29.4	30.5	39.2	56.6	74.8	79.1	83.1	78.2	65.8	52.6	40.7	32.8		
		MCWB	28.3	28.5	34.3	46.1	58.1	64.7	68.3	65.3	56.9	47.0	39.3	31.7		
	10%	DB	25.7	27.1	35.9	52.0	70.0	75.2	79.5	74.5	61.8	49.2	36.5	30.3		
		MCWB	24.4	25.0	32.4	43.1	55.5	62.9	66.7	63.9	55.5	45.7	34.9	29.2		
	0.4%	WB	33.3	33.5	41.2	55.8	65.0	69.6	72.7	71.4	62.5	52.6	46.7	35.7		
		MCDB	34.0	34.5	48.9	68.0	79.5	80.7	84.1	85.1	71.3	59.1	49.0	35.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	31.5	31.5	37.4	51.4	61.7	67.9	70.9	69.3	60.1	50.1	42.4	33.3		
		MCDB	32.3	33.1	41.3	60.1	75.7	78.7	83.2	80.7	67.2	55.8	44.1	33.9		
	5%	WB	28.6	28.5	34.9	47.4	59.3	66.3	69.2	67.3	58.3	47.7	39.1	31.7		
		MCDB	29.4	30.3	38.4	54.5	71.9	76.7	80.5	76.1	63.5	51.7	40.6	32.7		
	10%	WB	24.6	25.0	32.9	44.5	57.1	64.2	67.6	64.9	56.3	45.6	35.0	29.0		
		MCDB	25.9	26.8	35.7	51.1	67.9	73.5	77.5	72.3	61.0	49.2	36.1	30.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	12.5	13.1	15.0	19.5	17.2	17.5	15.9	12.6	8.3	7.0	8.7		
		MCDBR	8.9	9.4	15.7	21.8	25.4	22.5	22.4	21.3	17.8	14.4	8.6	6.2		
		MCWBR	8.4	8.0	10.4	12.4	11.2	9.5	8.9	9.4	8.9	8.6	7.9	6.5		
		MCDBR	9.0	9.1	13.7	19.6	23.2	20.7	20.4	19.9	14.8	12.5	9.1	6.5		
	5% WB	MCWBR	8.8	8.0	9.8	11.7	11.3	10.2	9.0	9.5	8.5	8.4	8.3	6.8		
(40) Clear-Sky Solar Irradiance	taub		0.290	0.316	0.371	0.401	0.392	0.388	0.411	0.424	0.377	0.347	0.308	0.285		
	taud		1.999	2.006	1.938	2.086	2.302	2.369	2.311	2.247	2.341	2.274	2.095	1.970		
	Ebn at Noon		160	215	235	253	267	270	259	245	239	209	154	120		
	Edh at Noon		19	30	42	43	37	36	37	37	29	23	17	14		
(44) All-Sky Solar Radiation	RadAvg		165	442	844	1252	1603	1704	1716	1276	752	319	136	90		
	RadStd		26	73	107	170	179	150	180	134	73	46	27	19		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page